



UNIVERSITY OF
LEICESTER

**School of Computing and Mathematical
Sciences**

CO7201 Individual Project

Interim Report

**CulinaAI: An AI-Assisted Recipe Generation
and Meal Personalisation System**

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Declaration

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Table of Contents

1. Project Update	3
2. Completed, Ongoing and Pending Tasks	3
2.1 Completed Tasks	3
2.2 Ongoing Tasks	3
2.3 Pending Tasks	3
3. Updated Time Plan and Gantt Chart	6
3.1 Specific Technical Challenges and Solutions	6
3.2 Comparison with Existing Recipe Platforms	6
3.3 Compact Gantt Chart	7
3.4 Updated Planning Notes	7
Appendix A: Visual Confirmation of Task Completion	8
Appendix B: System Design Diagrams	11
Appendix C: Codebase and Database Evidence	13
C.1 Project/GitLab Evidence	13
C.2 Key Codebase Evidence	13
Appendix D: GitLab and Testing Evidence	13
D.1 Manual Testing Summary	13
D.2. Backend/API Testing Evidence	14
D.3 Known Bugs and Fixes.....	14

1. Project Update

CulinaAI has progressed from the preliminary report proposal into a working Django-based prototype. The project currently includes user authentication, dashboard, AI-assisted recipe generation, recipe result display, recipe history, saved recipes, smart pantry, shopping list, AI cooking assistant and interactive cooking mode. The core aim from the preliminary report remains the same: to build an AI-assisted meal personalisation system that generates recipe suggestions based on user ingredients, dietary preferences, allergies, cooking time and other cooking constraints.

At the interim stage, the project is approximately 75-80% complete in terms of implementation. Most essential requirements have been implemented and several recommended or optional features have also been completed. The remaining work is mainly focused on automated testing, documentation, final screenshots, GitLab evidence, bug fixing and preparation for the principal marker interview.

Compared with ordinary recipe websites, CulinaAI is not limited to displaying fixed recipes. It generates personalised recipes from user constraints, stores user-specific outputs, supports pantry-based planning, provides shopping list support and allows users to ask recipe-related cooking questions. The main technical contribution so far is the integration of AI recipe generation with a structured Django database workflow and user-specific feature management.

2. Completed, Ongoing and Pending Tasks

2.1 Completed Tasks

- User authentication: registration, login, logout, password reset and profile functionality have been implemented. Authenticated pages such as dashboard, saved recipes and pantry are protected.
- Dashboard: the user dashboard displays activity summaries including saved recipes, AI generations, favourite recipes, feedback and recent recipe activity.
- AI recipe generation: users can enter ingredients, cuisine, meal type, dietary preference, allergies, cooking time, servings and other preferences to generate personalised recipes.
- Recipe result display and storage: generated recipes are shown in a structured format and saved recipe records are linked to the logged-in user.
- Smart pantry: users can add pantry ingredients with quantity, unit, expiry date, notes and availability status to support recipe planning.
- Shopping list: recipe ingredients can be extracted and compared with pantry items to identify missing items.
- AI cooking assistant and cooking mode: users can ask recipe-specific cooking questions and follow recipes step by step.
- Admin/database management: Django Admin and backend models are available for managing users, recipes, pantry items, ratings, feedback and related records.

2.2 Ongoing Tasks

- Automated testing: manual testing has been carried out, but automated tests still need to be expanded for authentication, recipe generation, saved recipes, pantry and validation-related logic.
- Evidence collection: screenshots, codebase evidence, GitLab commit history and testing evidence are being organised for the interim submission and principal marker interview.
- Final UI polishing: minor layout consistency improvements and small bug fixes are still ongoing.
- Principal marker preparation: a short demo flow is being prepared to explain the motivation, requirements, architecture, implemented features, technical challenges and remaining work.

2.3 Pending Tasks

- Complete formal automated tests for the main workflows.
- Finalise manual testing table and screenshots.
- Add or clean system diagrams for architecture, database design and user flow.
- Prepare final report template and implementation notes.
- Perform final GitLab/code cleanup before final submission.
- Continue final bug fixing and viva/principal marker preparation.

The table below summarizes progress against the preliminary requirements using exact dates, implemented work and the main technical issue addressed in each period.

Week	Dates	Progress, Tasks and Technical Challenges	Status
Week 1	1 July - 5 July 2026	Set up the Django project structure, organised backend/frontend folders, configured static files and created the base homepage, navigation and layout. The challenge was keeping the structure close to the README while following Django template/static-file organisation.	Completed
Week 2	6 July - 10 July 2026	Implemented signup, login, logout, password reset and profile functionality. Developed the dashboard and protected authenticated pages. The challenge was ensuring user-specific access so users only see their own dashboard, recipes and pantry information.	Completed
Week 3	11 July - 15 July 2026	Developed recipe models, recipe generation form, backend validation and AI recipe generation service. The challenge was collecting enough information for personalisation without making the interface confusing.	Completed
Week 4	16 July - 20 July 2026	Implemented recipe result display, automatic recipe history, saved recipes and favourite workflow. Added recipe modification and improved recipe storage. The challenge was separating generated history from intentionally saved recipes while linking all records to the correct user.	Completed
Week 5	21 July - 24 July 2026	Implemented smart pantry, shopping list, AI cooking assistant, interactive cooking mode, admin/database evidence and interim report preparation. The challenge was connecting advanced features without increasing scope too far before the interim deadline.	Completed/Ongoing

The main achievements so far are:

- A working full-stack prototype has been developed.
- The system supports account-based recipe generation and recipe management.
- AI generation is integrated into the Django workflow.
- User-specific data is stored and managed through database models.
- Smart pantry and shopping list features extend the project beyond a basic recipe website.
- AI cooking assistant and cooking mode improve user interaction after recipe generation.

The main technical risks are AI output inconsistency, insufficient automated testing and possible scope expansion. These are being managed by keeping core features stable, recording evidence, avoiding unnecessary new features and prioritising testing before final submission.

3. Updated Time Plan and Gantt Chart

The updated plan below shows the remaining work from the interim submission to the final project deadline. Feature development is now being reduced so that the remaining time can focus on testing, evidence collection, final report writing, bug fixing and viva preparation.

3.1 Specific Technical Challenges and Solutions

Technical Feature	Specific Technical Difficulty	Solution Implemented
AI recipe validation	AI-generated recipes can be inconsistent, incomplete or may conflict with user restrictions.	Added validation checks for recipe structure, ingredient relevance, allergy terms, diet preference, cooking time and difficulty level.
Smart pantry	Pantry items are more complex than normal ingredients because they include quantity, unit, expiry date, notes and availability.	Created user-linked pantry records and added expiry/availability logic so expired or unavailable items are handled separately.
Dashboard insights	The dashboard needs to summarise data from different features such as saved recipes, favourites, feedback and generation history.	Used user-specific database queries to calculate activity summaries and display them as dashboard cards and insights.
Interactive cooking mode	Recipe instructions need to be followed step by step, not shown only as one long text block.	Processed recipe instructions into separate cooking steps so users can follow the recipe more practically.
Voice input in recipe form	Spoken ingredients must be converted into text and inserted into the correct form field without affecting manual typing.	Added microphone input as an enhancement while keeping normal typed input available.
Shopping list support	Recipe ingredients are text-based and may not directly match pantry item names.	Added logic to extract ingredient lines and compare them with pantry items to identify available and missing ingredients.

3.2 Comparison with Existing Recipe Platforms

Area	Existing Recipe Platforms	CulinaAI Extra/Unique Addition
Recipe source	Mainly fixed recipe collections or searchable recipe databases.	Generates personalised recipes dynamically using user inputs.
Ingredient support	Some platforms allow ingredient-based search.	Uses manual ingredients and pantry items with quantity, expiry and availability logic.
Cooking guidance	Some apps provide step-by-step cooking instructions.	Adds interactive cooking mode connected to the generated/saved recipe.
Personalisation	Usually based on filters such as cuisine, diet or ingredients.	Combines ingredients, diet, allergies, time, servings, difficulty, spice level and budget.
AI safety/quality	Most recipe sites display prepared recipes, so validation is not usually visible.	Adds recipe validation checks for structure, ingredient relevance, diet, allergy conflicts and risk level.
User insights	Basic saved recipes or account features may exist.	Dashboard summarises saved recipes, favourites, feedback, modifications and activity. Adds microphone input for entering ingredients/preferences more easily.
Voice input	Not always part of normal recipe forms.	

Pantry/shopping link	Some platforms support meal plans or shopping lists.	Compares recipe ingredients with pantry items to identify available and missing items.
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3.3 Compact Gantt Chart

Task / Period	24 Jul	25-31 Jul	1-10 Aug	11-17 Aug	18-24 Aug	25-31 Aug	1-4 Sep
Interim report submission							
Principal marker demo and GitLab check							
Automated testing							
Manual testing evidence							
Final report template and literature notes							
Architecture/database/workflow diagrams							
UI polish, bug fixing and documentation							
Evaluation and viva preparation							
Final code cleanup and submission checks							

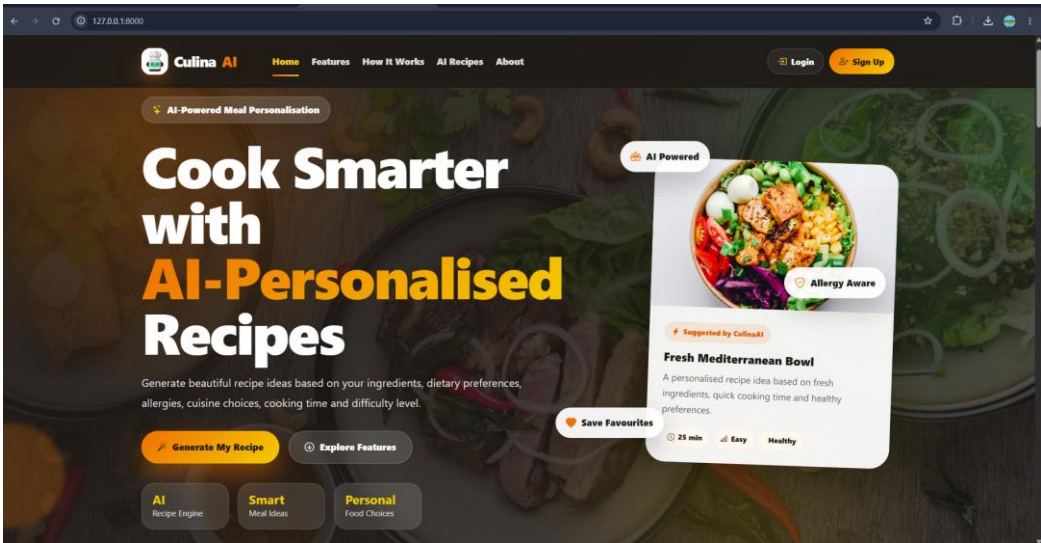
3.4 Updated Planning Notes

- The principal marker period is prioritised immediately after submission, with the latest GitLab commit and a stable prototype demo prepared by 31 July 2026.
- From 1 August to 10 August, the focus moves to automated tests and manual testing evidence for authentication, recipe generation, saved recipes, pantry and cooking features.
- From 11 August onward, the project focus shifts from adding features to final report preparation, diagrams, bug fixing, evaluation and viva readiness.
- No major new feature is planned before final submission unless it is required to fix an existing workflow or improve project reliability.

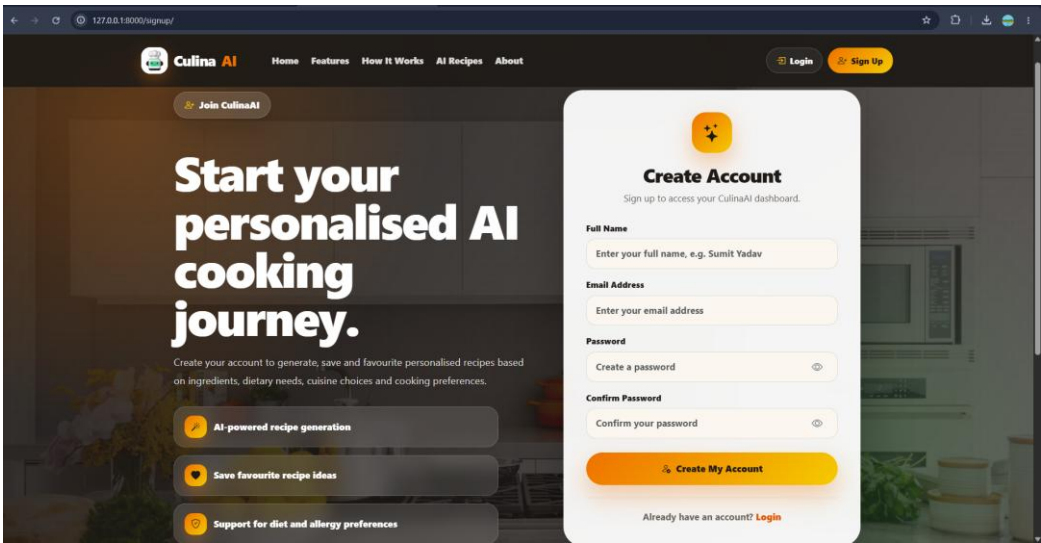
The remaining plan has been adjusted from the preliminary report because the core implementation progressed faster than expected, allowing pantry, shopping list and cooking assistant features to be developed before the interim deadline. The priority for the next stage is not adding major features, but testing, evidence collection, final documentation and preparing a reliable demonstration.

Appendix A: Visual Confirmation of Task Completion

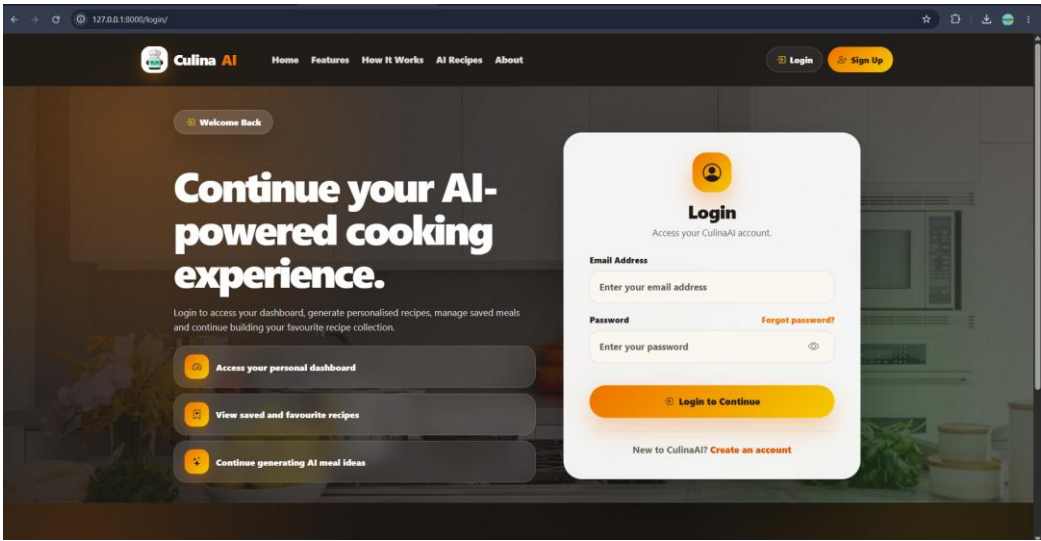
A.1 Homepage



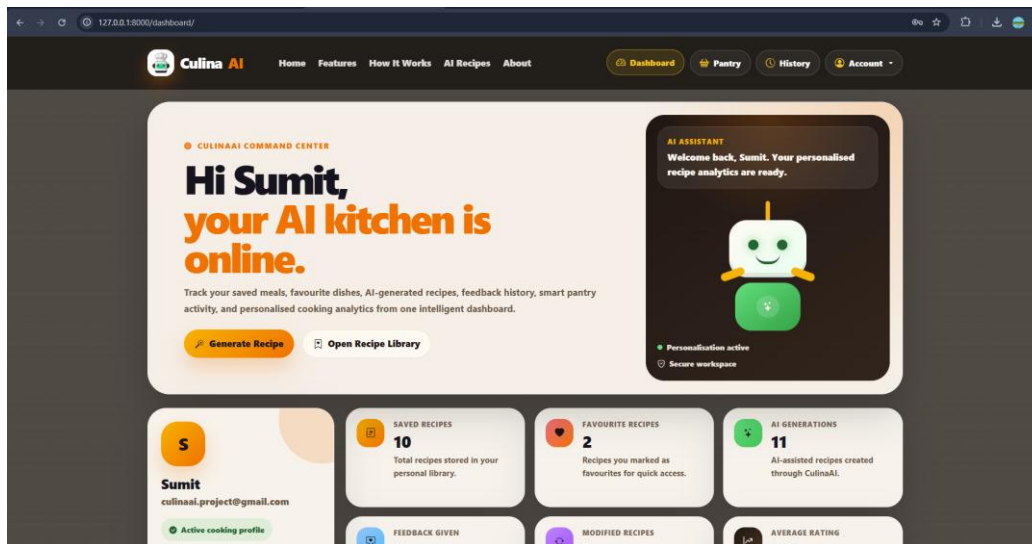
A.2 Signup Page



A.3 Login Page



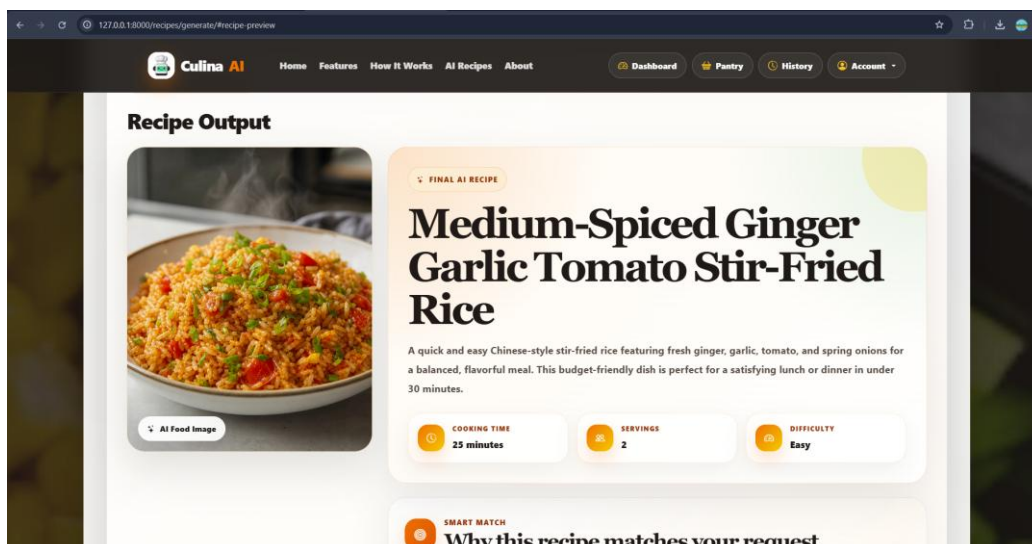
A.4 User Dashboard



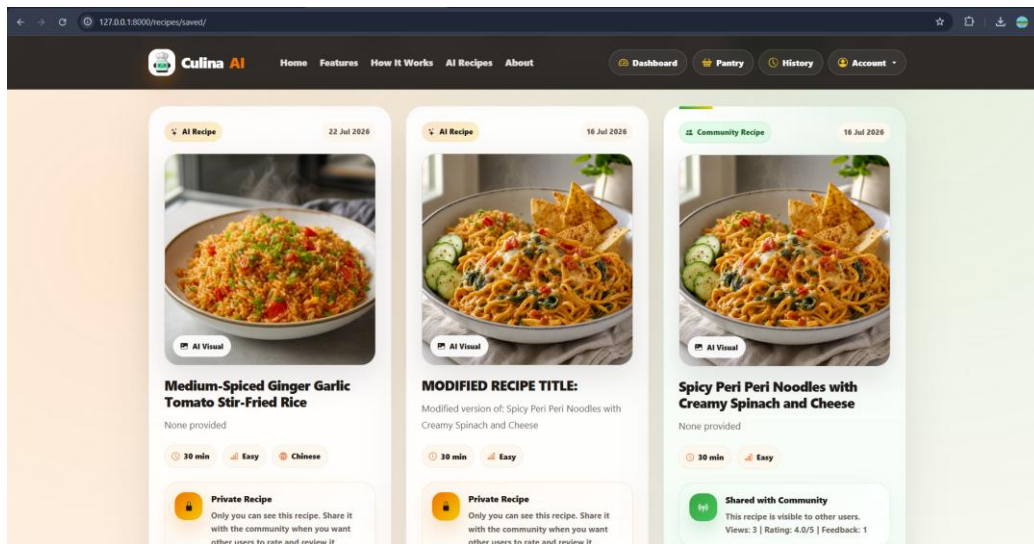
A.5 Recipe Generation Form

The 'Generate Recipe' form is titled 'PERSONALISE YOUR MEAL'. It includes a 'Turn your ingredients into a personalised recipe.' sidebar with options to 'Use ingredients you already have' and 'Include allergies and dietary restrictions'. The main form has three sections: 'MAIN RECIPE DETAILS' with 'Available Ingredients' (example: chicken, rice, tomatoes, garlic, spinach) and a 'Speak' button; 'COOKING SETUP' with 'Cooking Time minutes' (30), 'Number of Servings' (2), and 'Difficulty' (Easy); and 'TASTE AND GOAL' with 'Spice Level', 'Budget Level', and 'Nutrition Goal'.

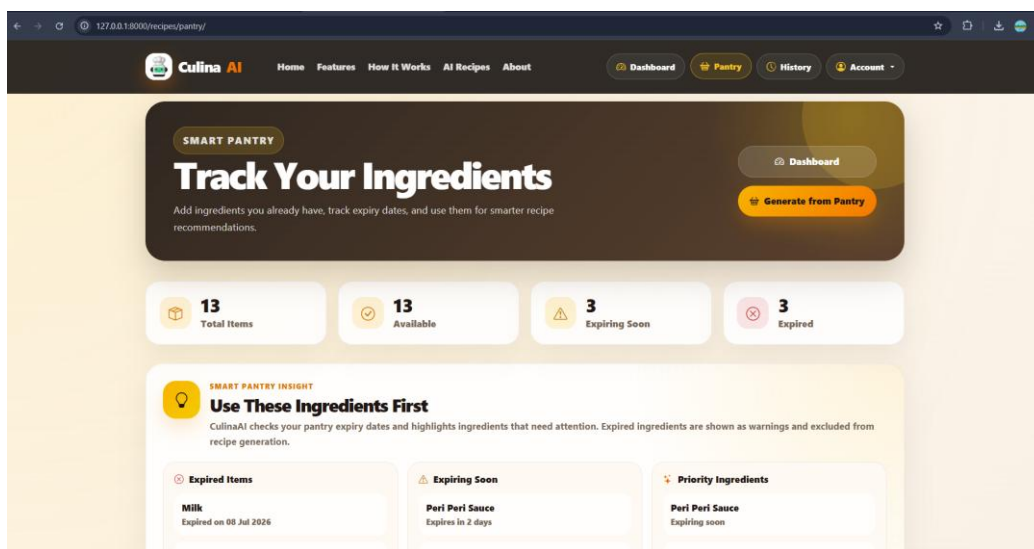
A.6 AI Recipe Result



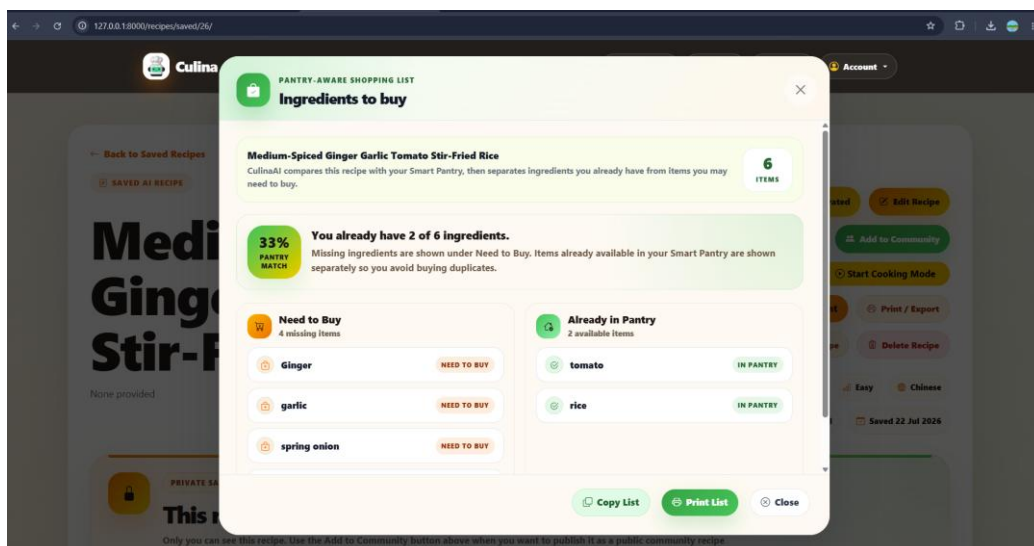
A.7 Saved Recipes



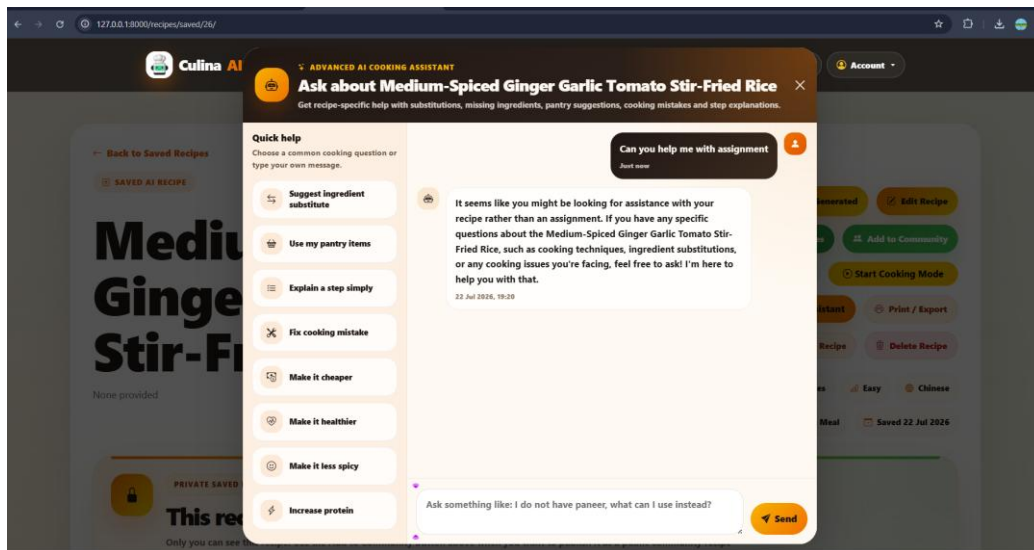
A.8 Smart Pantry



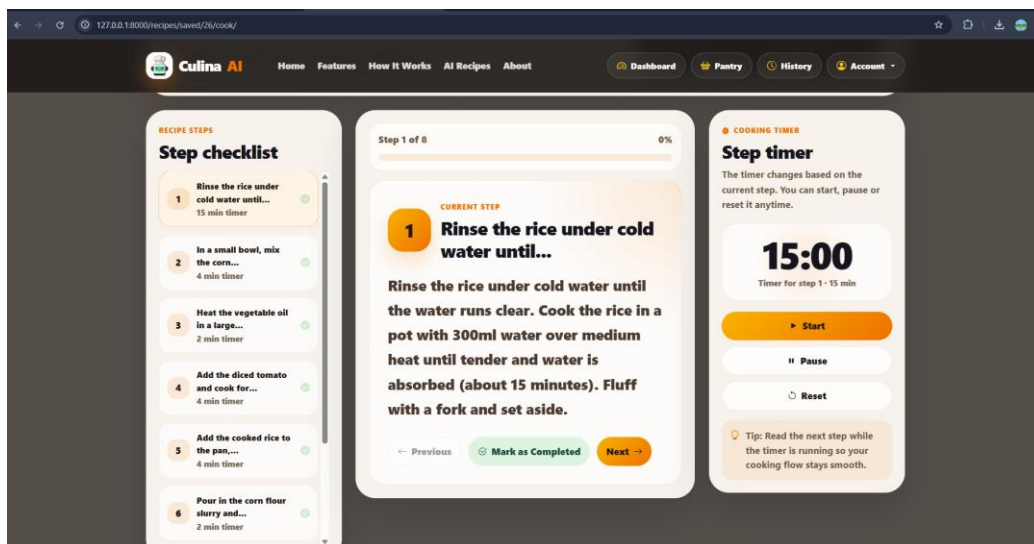
A.9 Shopping List



A.10 AI Cooking Assistant

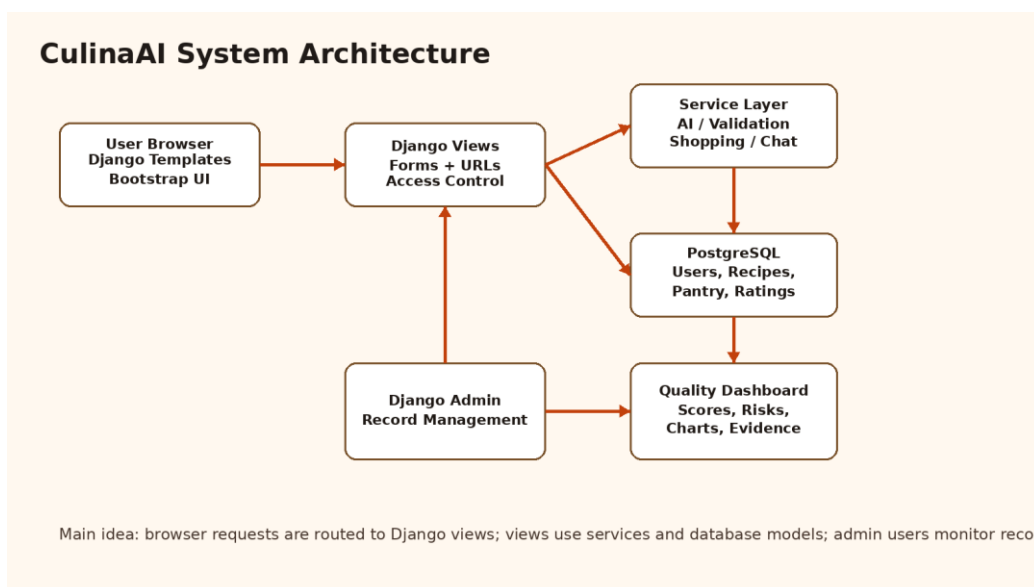


A.11 Interactive Cooking Mode

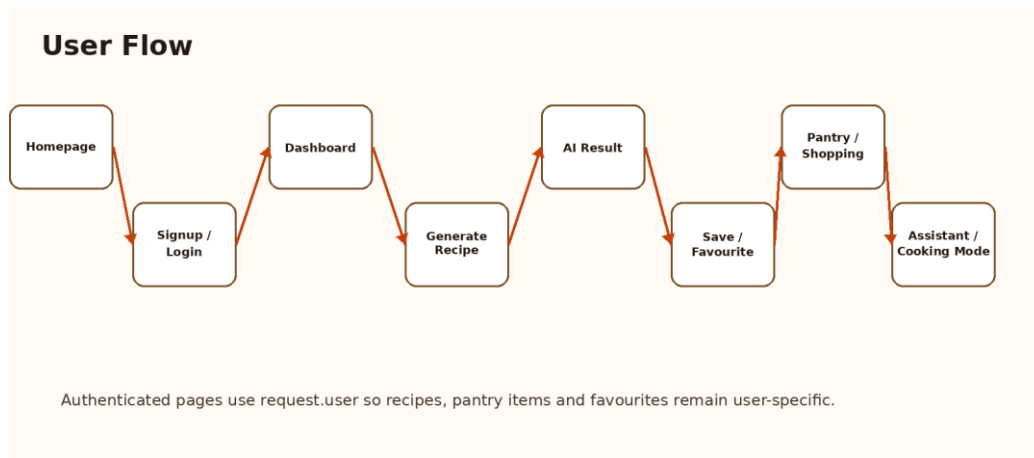


Appendix B: System Design Diagrams

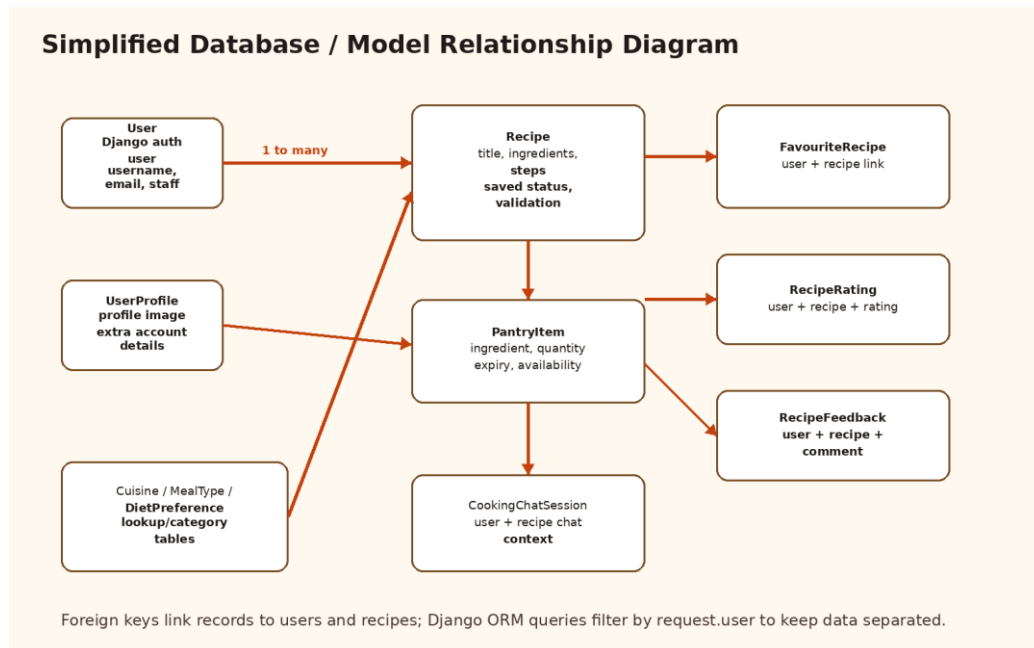
B.1 System Architecture Diagram



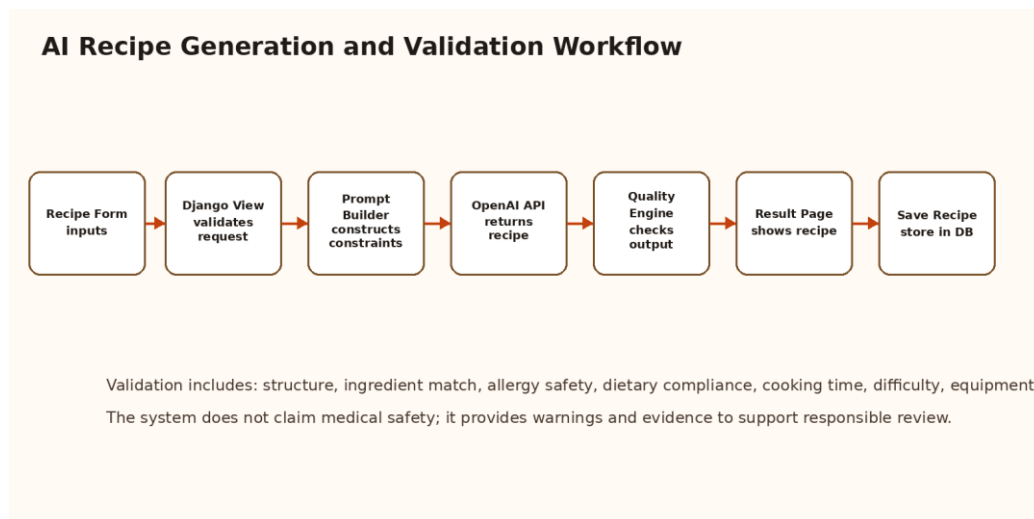
B.2 User Flow Diagram



B.3 Simplified Database/Model Relationship Diagram

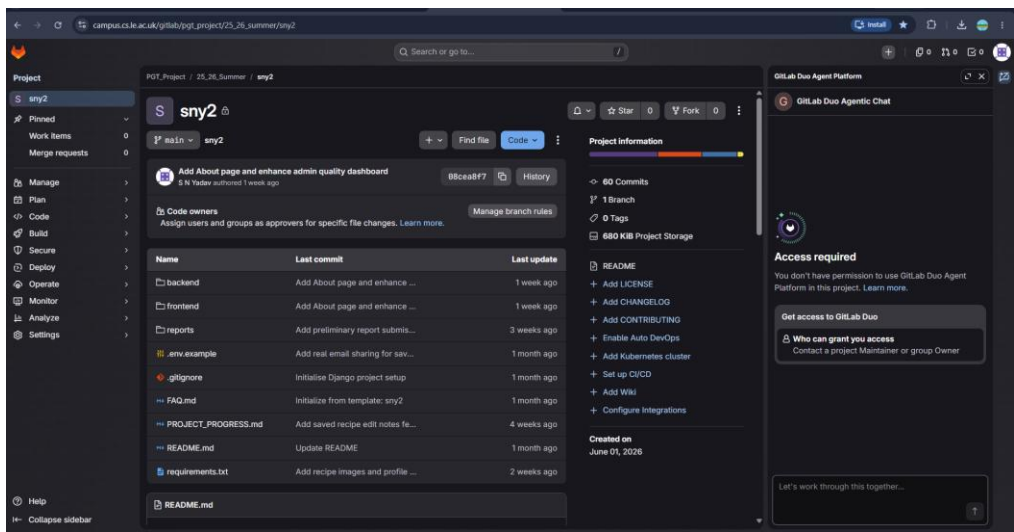


B.4 AI Recipe Generation and Validation Workflow



Appendix C: Codebase and Database Evidence

C.1 Project/GitLab Evidence



C.2 Key Codebase Evidence

- Models: Recipe, RecipeHistory, PantryItem, UserProfile, RecipeRating and RecipeFeedback.
- Key views/services: AI recipe generation, saved recipes, pantry, shopping list, cooking assistant and cooking mode.
- Database evidence to be used in the principal marker interview: Django Admin records and SQL checks for users, recipes, validation scores and user-linked records.

Appendix D: GitLab and Testing Evidence

D.1 Manual Testing Summary

Test Case	Input/Action	Expected Result	Status
User signup	Valid username/email/password	Account created	Pass
User login	Valid credentials	Dashboard opened	Pass
Generate recipe	Ingredients + preferences	Recipe generated	Pass
Save recipe	Click Save	Recipe appears in saved list	Pass
Pantry item	Add ingredient with expiry	Item appears in pantry	Pass
Shopping list	Open saved recipe and generate list	Missing/available ingredients displayed	Pass
AI assistant	Ask recipe-specific question	Relevant cooking response shown	Pass
Cooking mode	Open recipe in cooking mode	Step-by-step mode displayed	Pass

D2. Backend/API Testing Evidence

Figure D.2: Backend/API testing evidence showing successful request handling for the CulinaAI project.

The screenshot shows a VS Code editor with a file explorer on the left displaying the project structure for 'SNY2'. The file explorer shows a 'backend' directory containing 'accounts', 'config', 'dashboard', 'recipes', and 'tests.py'. The 'tests.py' file is open in the editor, showing a Python class 'BasicProjectTests' with a method 'test_user_can_login'. The terminal at the bottom shows the output of running 'python manage.py test' in a PowerShell prompt. The output indicates that 3 tests were found and passed successfully, with a total execution time of 0.687s. The status bar at the bottom shows the current file is 'tests.py' at line 31, column 42, with 4 spaces, UTF-8 encoding, and CRLF line endings. The Python version is 3.14.6.final.0.

```
backend > accounts > tests.py > BasicProjectTests > test_user_can_login
5 class BasicProjectTests(TestCase):
19 def test_user_can_login(self):
29
30
31 self.assertTrue(login_successful)
```

```
(venv) PS C:\Users\yadav\Documents\C07201_Project\sny2\backend> python manage.py test
Found 0 test(s).
System check identified no issues (0 silenced).

-----
Ran 0 tests in 0.000s

NO TESTS RAN
(venv) PS C:\Users\yadav\Documents\C07201_Project\sny2\backend> python manage.py test
Found 3 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
...
-----
Ran 3 tests in 0.687s

OK
Destroying test database for alias 'default'...
(venv) PS C:\Users\yadav\Documents\C07201_Project\sny2\backend>
```

D.3 Known Bugs and Fixes

Issue	Cause	Fix	Status
Footer alignment issue	Nested footer column/layout conflict	Rebuilt role-based footer layout	Fixed
Missing community recipe URL error	Invalid URL name in base template	Replaced with homepage section link	Fixed
Shopping list print/copy issue	Modal and JavaScript behaviour	Updated script and print styling	Fixed